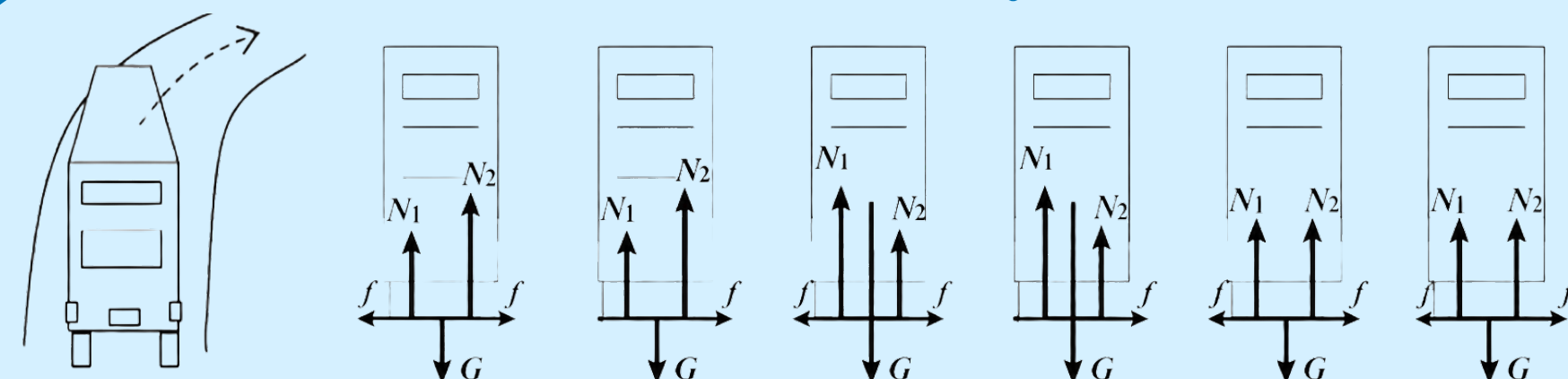


K-12 Sample:

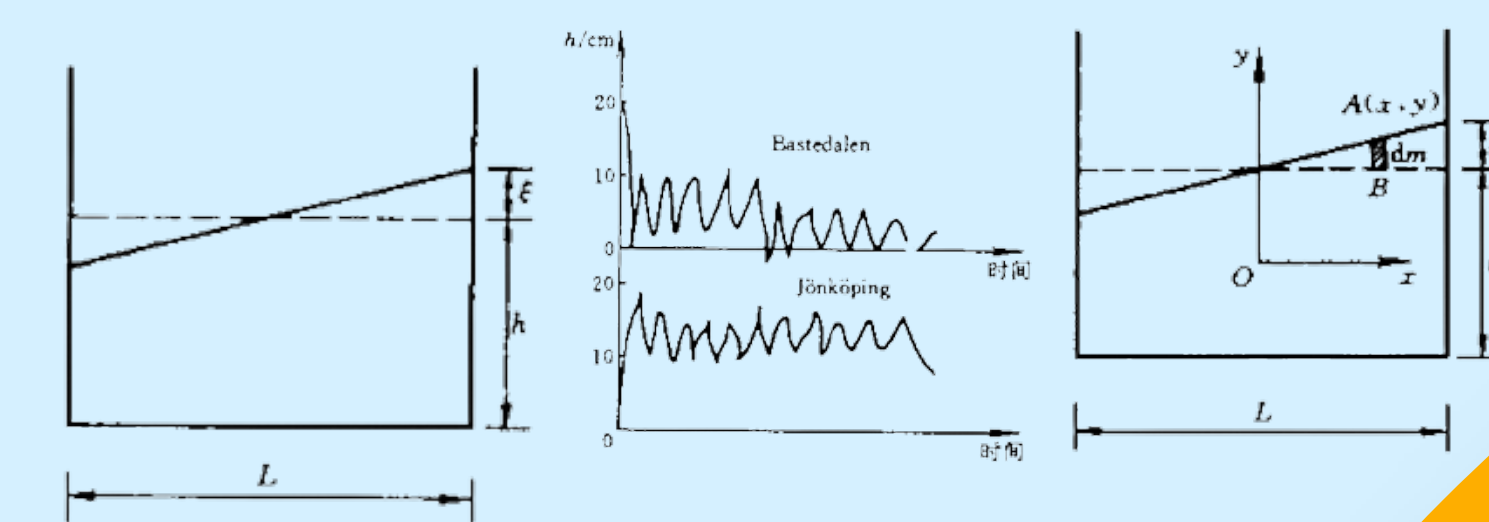


The figure shows a bus turning right on a flat road, viewed from the rear. Which of the following free-body diagrams is most correct?

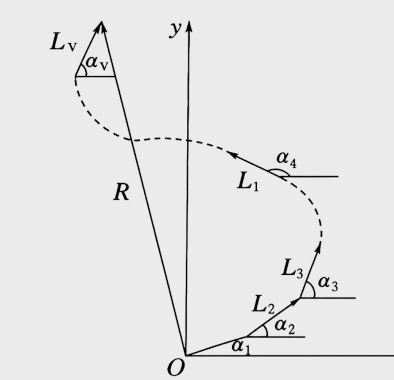
Mechanics

Olympiad Sample:

In some lakes, a peculiar phenomenon called seiche (water oscillation) can be observed. This usually occurs in shallow, long and narrow lakes, where the whole body of water appears to move, similar to how you would carry a cup of coffee to a guest. This should not be confused with surface waves on the lake. We model this using a rectangular container of length L and water depth h . Initially, the water surface makes a small angle with the horizontal plane. During oscillation, the surface remains flat and rotates about a horizontal axis located at the midpoint of the container's length. 1. Establish a model for the water motion and derive an expression for the oscillation period T . The initial condition is shown in the Figure. Assume $\xi \ll h$. 2. Two tables show the oscillation times for different water depths in containers of two different lengths. Use appropriate methods to check whether your derived formula matches the experimental data well, and comment on the applicability of this physical model. 3. Figure shows measurements taken on Lake Vattern in Sweden. The two curves represent measurements from the northern end (Basteden) and southern end (Jönköping). The lake has a length of 123km and an average depth of 50m. What time scale should be used in this case?



K-12 Sample:

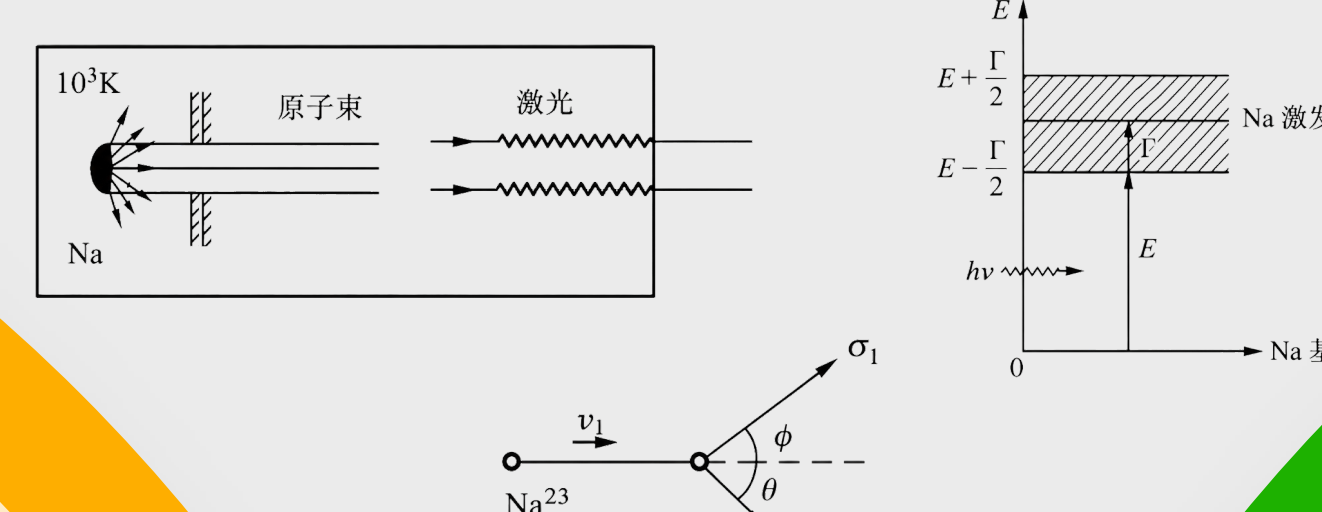


A sailor takes steps of length L in random directions. After N steps, on average, how far is he from the hotel?

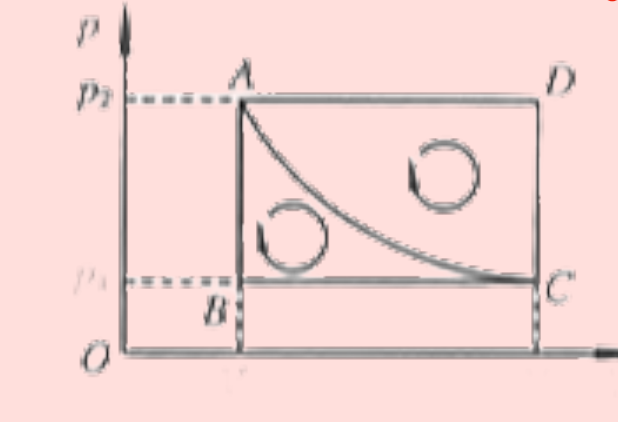
Modern Physics

Olympiad Sample:

Laser cooling of atoms: A beam of Na^{23} atoms (obtained by evaporation at 10^3 K) in a vacuum chamber is irradiated head-on by a high-intensity laser. The laser frequency is selected so that sodium atoms with velocity v_0 resonantly absorb photons, transition to the first excited state with energy E and width Γ after absorption, with velocity change Δv_1 , and then emit photons to return to the ground state, with velocity change $\Delta v'$ and deflection angle α . The process can be repeated multiple times, and the laser frequency needs to be changed to continue decelerating after the velocity decreases to a certain Δv . Given $m \approx 23m_p$ (m_p is proton mass), h (Planck constant), k (Boltzmann constant), c (speed of light). Find: 1. Laser frequency u for atoms with kinetic energy equal to the average kinetic energy and velocity change Δv_1 in the first absorption; 2. Velocity interval Δv_0 of atoms that can absorb photons of this frequency; 3. Maximum deflection angle α_{max} of atoms after one emission; 4. Maximum velocity reduction Δv of atoms when frequency is constant; 5. Number of absorptions N required to decelerate from initial velocity v_0 to nearly zero; 6. Time required for deceleration and distance Δs when ignoring emission velocity change.



K-12 Sample:

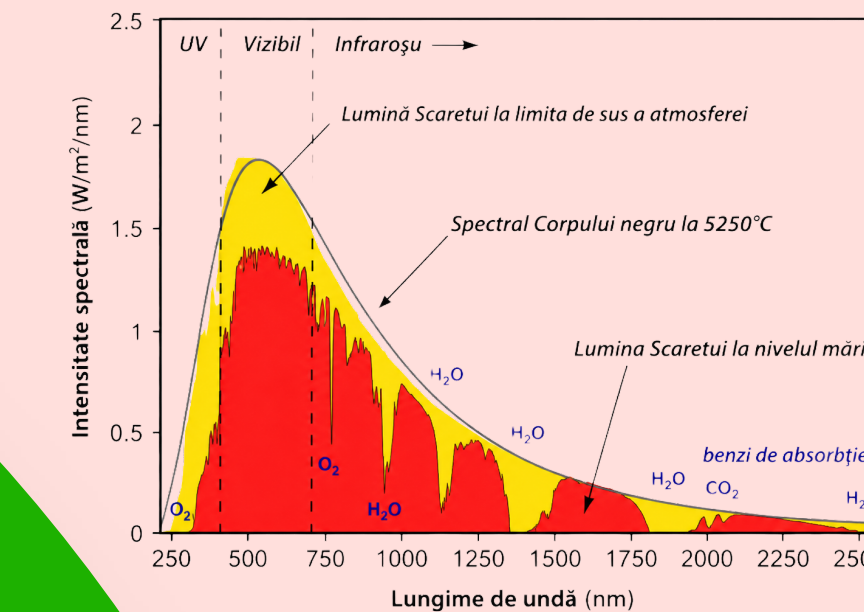


A monatomic ideal gas undergoes two cyclic processes shown: ACBA and ADCA. AC is an adiabat, and $T_D = 16T_B$. Find the ratio of the efficiencies $\eta_{ACDA} : \eta_{ACBA}$.

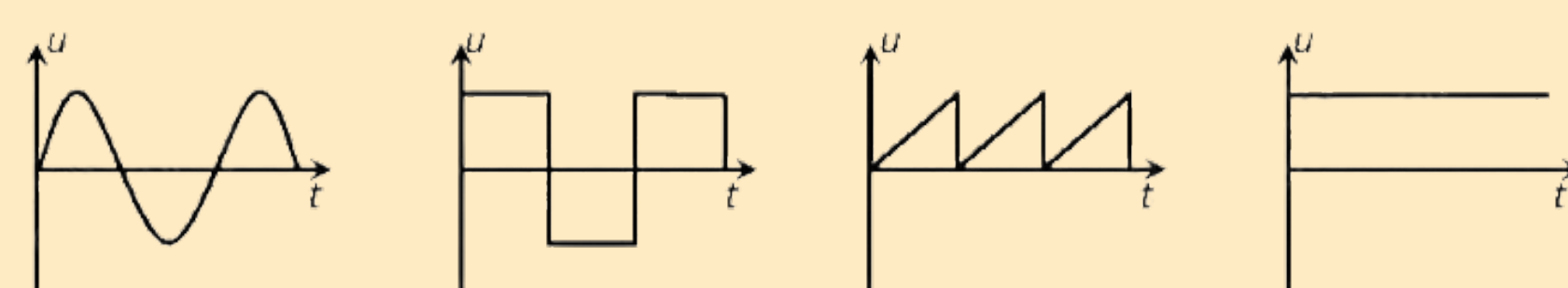
Thermodynamics

Olympiad Sample:

Terrestrial climate modeling Climate change and global warming, the life of humans in new climatic conditions becomes topics of public interest in recent decades. On a cosmic scale, the only phenomena occurring in Earth's energy balance are absorption and emission of radiation. One can say that the earth's climate condition depends on delicate balance between the energy that our planet receives for the formidable energy source that is the sun and the energy that Earth radiates into space. Consider that solar constant for the radiation coming from the Sun towards the Earth is $w_s = 1370 \text{ W} \cdot \text{m}^{-2}$. Stefan-Boltzmann law states that $\epsilon(T) = \sigma T^4$, where $\sigma = 5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$. Task 1: 1.a. Expression of average temperature T_p that the Earth's surface should have, in accordance with simple modeling used. 1.b. Numerical value of the average temperature T_p . Task 2: Consider the Earth's Albedo $A = 0.3$. 2.a. Expression of average temperature T_p of the Earth's surface considering the Albedo. 2.b. Numerical value T_p of the average temperature. Task 3: 3.a. Expression of average temperature T_p of the Earth's surface considering the atmosphere's greenhouse effect. 3.b. Numerical value T_p of the average temperature based on given atmospheric transmission and emission coefficients.



K-12 Sample:

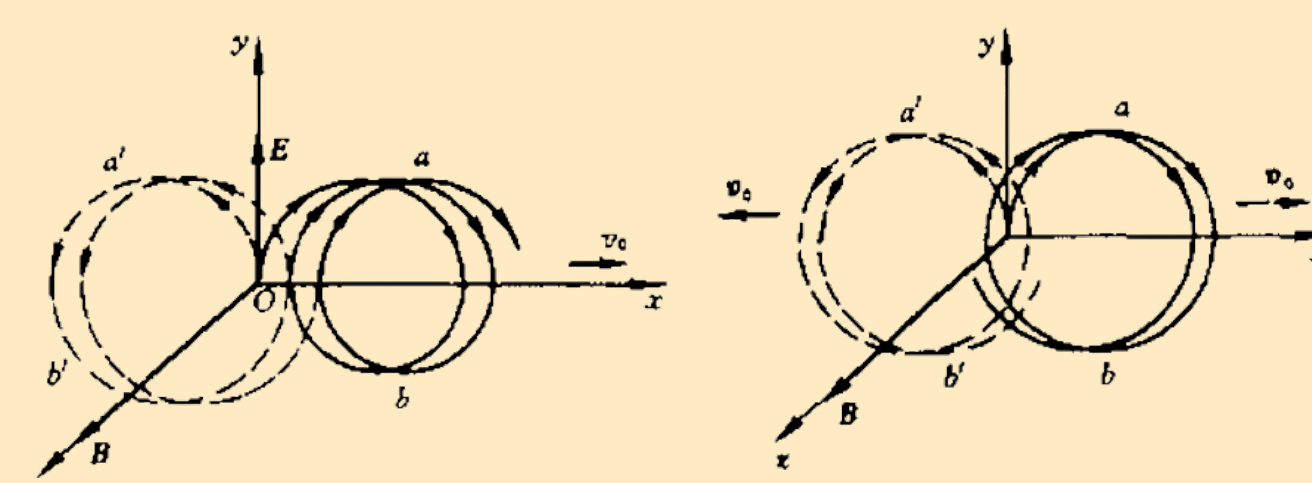


As shown in the figure, which of the following graphs shows the variation of the scanning voltage u with time t in a common oscilloscope?

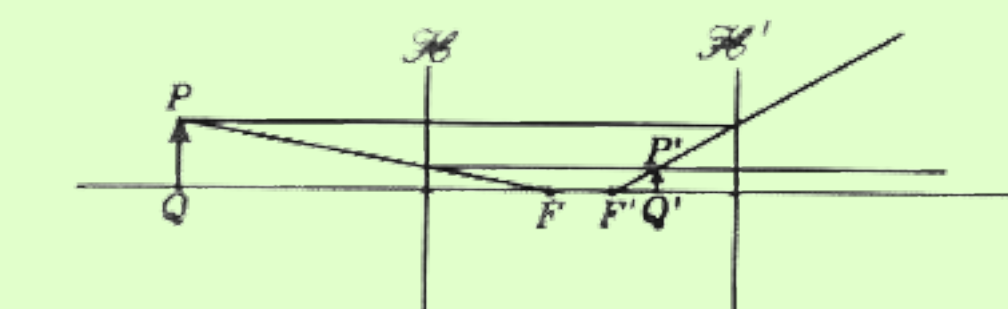
Electromagnetism

Olympiad Sample:

A particle source located at the origin O emits identical charged particles of mass m and charge q with initial velocity v_0 along the positive y -axis. The interaction between particles and gravity are negligible. 1. If there is a uniform magnetic field B along the positive z -axis in the region where the particles move, the charged particles will perform circular motion in the xy -plane. If a uniform weak electric field E along the positive y -axis is added, the circular orbit will drift along the positive or negative x -direction. For the cases $q>0$ and $q<0$, qualitatively determine the direction of this drift, sketch the trajectory, and calculate the corresponding drift velocity (appropriate approximations are allowed). 2. If the electric field E in the first part is not necessarily weak but still uniform and along the positive y -axis, derive rigorously the time dependence of the velocity components v_x and v_y along the x - and y -directions. Take the moment of emission from the origin as the initial time $t = 0$. 3. Remove the electric field in parts 1 and 2. Suppose the magnetic field is non-uniform, with a small linear gradient $\partial B/\partial y$ equal to a positive constant. Then the charged particles will also drift along the x -direction. Qualitatively determine the drift direction for positively and negatively charged particles, and sketch the trajectories.



K-12 Sample:



Use graphical method to find the image of points P and Q in the given figure (incident rays travel from left to right).

Optics

Olympiad Sample:

As shown in the Figure, inside a uniform isotropic dielectric sphere of radius R and refractive index n , there is a point light source Q emitting isotropic natural light. The outside is a vacuum. (1) If the light is strictly imaged by the spherical surface at Q' , Q and Q' are called a pair of aplanatic points. Find the distances r and r' from Q and Q' to the center of the sphere. (2) Continuing from the previous question, when Q and Q' are aplanatic points, the light appearing to originate from Q' is not isotropic. Ignoring reflection, find the relative light intensity at point P (the distance PQ' is constant at ρ_0 , and the intensity at P_0 is I_0 ; the expression should not contain φ). (3) Continuing from above, reflection is inevitable. According to Fresnel's equations, for light traveling from a medium with index n_1 at an incident angle i_1 into a medium with index n_2 at a refracted angle i_2 , the amplitude transmission coefficients are: $t_s = \frac{2n_1 \cos i_1}{n_1 \cos i_1 + n_2 \cos i_2}$, $t_p = \frac{2n_1 \cos i_1}{n_2 \cos i_1 + n_1 \cos i_2}$. Still assuming the intensity at P_0 is I_0 and considering only a single refraction, find the light intensity and degree of polarization at P .

